

# Jiahao Shao

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jiahao-shao1.github.io

## Education

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**The Hong Kong University of Science and Technology**

2026 – Present

Ph.D. in Computer Science

Advisor: Yinghao Xu

**Zhejiang University**

2020 – 2024

B.E. in Automation

Advisor: Yiyi Liao

## Experiences

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**Ant Group**

2024 – 2026

Research Intern

Advisor: Yujun Shen

**Zhejiang University**

2023 – Present

Research Assistant

Advisor: Yiyi Liao

## Research Interests

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My long-term research goal is to build a general-purpose world model that can understand, simulate, and interact with the physical world. My research interests lie at the intersection of Computer Vision, Generative Modeling, and Embodied AI.

## Publications

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(\* Equal Contribution; † Equal Advising)

- [1] Jiahao Shao, Yuanbo Yang, Yiyi Liao, Yujun Shen, Ceyuan Yang†, Yinghao Xu†, “Thinking With Tools, Not With Pixels: Tool Calls as Text Scaffolds for Visual Reasoning”, *Under review*, 2026.
- [2] Yuanbo Yang, Jiahao Shao, Yiyi Liao, Jun Gao, “From a Spiral to a Sink: How Positional Encoding Shapes Attention Sinks in Diffusion Transformers”, *Under review*, 2026.
- [3] Zhen Xu\*, Hongyu Zhou\*, Sida Peng, Haotong Lin, Haoyu Guo, Jiahao Shao, Peishan Yang, Qinglin Yang, Sheng Miao, Xingyi He, Yifan Wang, Yue Wang, Ruizhen Hu, Yiyi Liao, Xiaowei Zhou, Hujun Bao, “Towards Depth Foundation Model: Recent Trends in Vision-Based Depth Estimation”, *Computational Visual Media*, 2026.

- [4] Chunyang Li\*, Yuanbo Yang\*, Jiahao Shao\*, Hongyu Zhou, Katja Schwarz, Yiyi Liao, “ReRoPE: Repurposing RoPE for Relative Camera Control”, *SIGGRAPH*, 2026.
- [5] Jiabao Wang, Hongyu Zhou, Yuanbo Yang, Jiahao Shao, Yiyi Liao, “The Constant Eye: Benchmarking and Bridging Appearance Robustness in Autonomous Driving”, arXiv preprint, 2026.
- [6] Hanlei Guo, Jiahao Shao, Xinya Chen, Xiyang Tan, Sheng Miao, Yujun Shen, Yiyi Liao, “ScenDi: 3D-to-2D Scene Diffusion Cascades for Urban Generation”, *Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [7] Yuanbo Yang\*, Jiahao Shao\*, Xinyang Li, Yujun Shen, Andreas Geiger, Yiyi Liao, “Prometheus: 3D-Aware Latent Diffusion Models for Feed-Forward Text-to-3D Scene Generation”, *Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [8] Jiahao Shao\*, Yuanbo Yang\*, Hongyu Zhou, Youmin Zhang, Yujun Shen, Vitor Guizilini, Yue Wang, Matteo Poggi, Yiyi Liao, “Learning Temporally Consistent Video Depth from Video Diffusion Priors”, *Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [9] Hongyu Zhou, Jiahao Shao, Lu Xu, Dongfeng Bai, Weichao Qiu, Bingbing Liu, Yue Wang, Andreas Geiger, Yiyi Liao, “HUGS: Holistic Urban 3D Scene Understanding via Gaussian Splatting”, *Computer Vision and Pattern Recognition (CVPR)*, 2024.

## Honors and Awards

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|-----------------------------------------------------------|-------------|
| Finalist Award, Mathematical Contest in Modeling (top 2%) | 2023        |
| Second-prize Scholarship of Zhejiang University           | 2020 – 2022 |

## Open Source Projects

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### **sjh-skills**

A collection of Claude Code skills for research workflow automation.

## Skills

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Python, C, PyTorch, PyTorch Lightning, MATLAB,  $\LaTeX$ , Linux